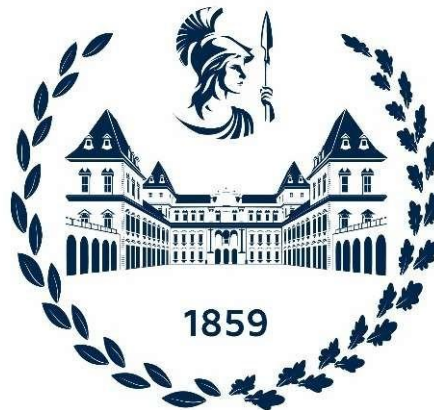


Caches: example (I)

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Scenario

Let us consider a processor connected to a 64KB memory and using a direct mapped cache composed of 16 lines, each storing 32 bytes of data.

Let assume that initially the 16 lines store the first 16 memory blocks (i.e., line 0 stores block 0, line 1 stores block 1, and so on).

Considering the following sequence of 12 memory accesses, we want to understand which of them causes a hit or miss.

Unaligned access is allowed.

Address	Block	Accessed line	H/M
0100 0000 0011 0011			
0100 0001 0001 1000			
0000 0100 1000 1110			
0010 0000 1011 1110			
0100 0000 1001 1111			
0100 0000 0011 0011			
0000 1010 0001 0011			
0000 1010 0101 0100			
0000 0011 0011 0100			
0000 0011 0011 0110			
0000 1000 1001 1000			
0000 0000 0001 1001			

Address	Block	Accessed line	H/M
0100000 0001 10011			
0100000 1000 11000			
0000010 0100 01110			
0010000 0101 11110			
0100000 0100 11111			
0100000 0001 10011			
0000101 0000 10011			
0000101 0010 10100			
0000001 1001 10100			
0000001 1001 10110			
0000100 0100 11000			
0000000 0000 11001			

- Each address corresponds to 16 bits
- The 5 least significant bits identify the byte within the block (OFFSET)
- The 11 most significant bits identify the block (TAG+INDEX)
- The 4 least significant bits among those identifying the block identify the line (INDEX).

Address	Block	Accessed line	H/M
0100000 0001 10011	513	1	M
0100000 1000 11000	520	8	M
0000010 0100 01110	36	4	M
0010000 0101 11110	133	5	M
0100000 0100 11111	516	4	M
0100000 0001 10011	513	1	H
0000101 0000 10011	80	0	M
0000101 0010 10100	82	2	M
0000001 1001 10100	25	9	M
0000001 1001 10110	25	9	H
0000100 0100 11000	68	4	M
0000000 0000 11001	0	0	M

- Each address corresponds to 16 bits
- The 5 least significant bits identify the byte within the block
- The 11 most significant bits identify the block
- The 4 least significant bits among those identifying the block identify the line.